

Ludvig Fellstrom

(305) 992-1971 | Inf33@cornell.edu | ludvigfellstrom.com | linkedin.com/in/ludvig-fellstrom

EDUCATION

Cornell University - College of Engineering

Ithaca, NY

B.S. Electrical and Computer Engineering | GPA: 3.6/4.0 | Expected May 2028

Relevant Coursework: Embedded Systems, Computer Systems Programming, Computer Architecture, Robotics, Circuits, Digital Logic

SKILLS

Languages & Platforms: C, C++, Python, Rust, MATLAB, SystemVerilog; Linux, Docker, Kubernetes, CI/CD

Embedded & Perception: RTOS (FreeRTOS), ROS 2, CAN, Ethernet, I2C, SPI, UART; PyTorch, TensorRT, MISRA C

Validation & Hardware: HIL/SIL, pytest, GDB, cocotb, Verilator, Altium, LTspice, PCBAs, oscilloscopes, logic analyzers

EXPERIENCE

IBM

Austin, TX | May 2026 - Aug 2026

Power Systems Development Intern

- Automated firmware validation through software-in-the-loop (SIL) testing and a Python-controlled HIL platform with robotic actuators, cameras, and sensors for 50+ Power servers; integrated ETM/Jazz test reporting.
- Debugged firmware and device-driver failures using schematics, oscilloscopes, logic analyzers, and telemetry; designed the test PCB and enclosure in Altium with watchdogs, interlocks, and safe sequencing.
- Built a dual-camera PyTorch perception pipeline (RTMPose, ArUco localization, CNN classification) to automatically verify hardware state across 50+ Power servers, replacing an estimated 40 hours/week of manual inspection.
- Deployed 3 Java/Quarkus microservices using REST APIs, ActiveMQ, PostgreSQL, and OpenShift; instrumented queue latency with Grafana/Prometheus and built Rust integration, load, and chaos tests.

Ghost Social

San Francisco, CA | Jun 2025 - Aug 2025

AI Engineering Intern

- Reduced dashboard latency from 45 seconds to 3 seconds through query redesign; diagnosed backend failures and shipped fixes through CI/CD.

LEADERSHIP

CUSail

Ithaca, NY | Sep 2024 - Present

Navigation Subteam Member

- Led embedded and electrical development for Cornell's autonomous sailboat; implemented C/C++ firmware and FreeRTOS tasks for PID control and wind, IMU, and GPS acquisition. Team won the 2025 International Robotic Sailing Competition.
- Designed and brought up an Altium PCB with buck converters and actuator routing; integrated UART and PWM peripherals and implemented timeout and recovery logic for CAN/XBee communications.
- Deployed YOLOv11x perception at approximately 93% mAP on NVIDIA Orin Nano using TensorRT and Docker; integrated inference with the ROS 2 autonomy stack.
- Validated firmware, electronics, and communications over 40+ field-test hours using ROS bags and GDB; reduced XBee latency by 30% and established static analysis and fuzz testing.

Cornell ECE 2400 - Computer Systems Programming

Ithaca, NY | Aug 2026 - Present

Teaching Assistant

- Supported 80+ students with C/C++ debugging, MISRA C coding practices, and development workflows; translated recurring issues into clearer technical guidance.

RESEARCH AND PROJECTS

Collective Embodied Intelligence Lab - Autonomous Buoy System

Ithaca, NY | Aug 2026 - Present

Undergraduate Researcher

- Integrated dissolved-oxygen, PT100 temperature, and fluorescence sensors with Arduino UNO Q over I2C/SPI for autonomous maritime sensing; processed and validated readings on Linux at 15-minute intervals.
- Designed a 10 W solar and 10,000 mAh LiPo system with duty-cycled sleep, sustaining 30+ days of unmanned open-water deployment.

Cornell High Frequency Trading Club

Ithaca, NY | Aug 2025 - Present

Founding Hardware Team Member

- Developed a SystemVerilog AXI4-Stream Ethernet parser extracting MAC addresses and EtherType from 64-bit data.
- Built C++ Ethernet/IPv4 golden models and cocotb RTL tests covering malformed traffic and resets; identified a parsing bug and contributed an accepted pull request.

ZT Group - Thermoelectric Wearables

Ithaca, NY | Aug 2025 - May 2026

Undergraduate Researcher

- Designed thermoelectric energy-harvesting PCBAs in Altium around an LTC3108, supercapacitor, and STM32; used LTspice for circuit analysis and developed C firmware with a custom temperature-sensor driver.
- Used MATLAB, DAQ measurements, and chamber testing to characterize thermal gradients and load response; applied DOE/SPC to manufacturing-related variability and diagnosed startup-voltage and PCB-degradation failures.